EX No:01

Date:

EDA using Auto Dataset

1.This exercise involves the Auto data set studied in the lab. Make sure that the missing values have been removed from the data.

LOAD LIBRARY:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import ISLP
from ISLP import load_data

auto = load_data('Auto')
auto.head()
```

OUTPUT:

name	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin
chevrolet chevelle malibu	18.0	8	307.0	130	3504	12.0	70	1
buick skylark 320	15.0	8	350.0	165	3693	11.5	70	1
plymouth satellite	18.0	8	318.0	150	3436	11.0	70	1
amc rebel sst	16.0	8	304.0	150	3433	12.0	70	1
ford torino	17.0	8	302.0	140	3449	10.5	70	1

(a) Which of the predictors are quantitative, and which are qualitative?

CODE:

```
quantitative = ['mpg', 'displacement', 'horsepower', 'weight', 'acceleration']
qualitative = ['cylinders', 'year', 'origin', 'name']
print("Quantitative predictors:", quantitative)
print("Qualitative predictors:", qualitative)
```

OUTPUT:

```
Quantitative predictors: ['mpg', 'displacement', 'horsepower', 'weight', 'acceleration']
Qualitative predictors: ['cylinders', 'year', 'origin', 'name']
```

(b) What is the range of each quantitative predictor? You can answer this using the range() function.

CODE:

```
ranges = auto[quantitative].apply(lambda x: (x.max(), x.min()))
print("Range of quantitative variables:")
print(ranges)
```

OUTPUT:

```
Range of quantitative variables:
    mpg  displacement  horsepower  weight  acceleration
0  46.6         455.0         230    5140          24.8
1   9.0          68.0          46    1613           8.0
```

(b) What is the mean and standard deviation of each quantitative predictor?

CODE:

```
mean_std = auto[quantitative].agg(['mean', 'std'])
print("Mean and standard deviation of quantitative predictors:")
print(mean_std)
```

OUTPUT:

```
Mean and standard deviation of quantitative predictors:
             mpg  displacement  horsepower       weight  acceleration
mean   23.445918    194.411990  104.469388  2977.584184     15.541327
std     7.805007    104.644004   38.491160   849.402560      2.758864
```

(c) Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?

CODE:

```
subset = auto.drop(auto.index[9:85])  # Python is 0-based
subset_range = subset[quantitative].apply(lambda x: (x.max(), x.min()))
subset_mean_std = subset[quantitative].agg(['mean', 'std'])
print("Range of quantitative predictors in the subset:")
print(subset_range)
print("Mean and standard deviation of quantitative predictors in the subset:")
print(subset_mean_std)
```

OUTPUT:

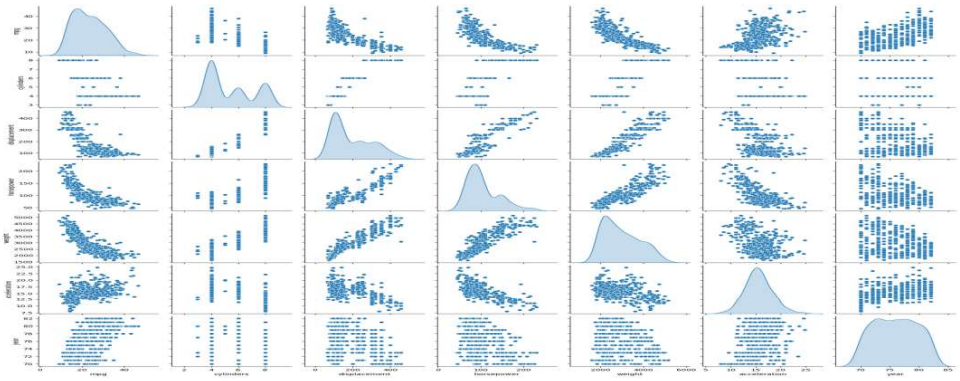
```
Range of quantitative predictors in the subset:
    mpg  displacement  horsepower  weight  acceleration
0  46.6         455.0         230    4952          24.6
1  11.0          68.0          46    1755           9.5
Mean and standard deviation of quantitative predictors in the subset:
             mpg  displacement  horsepower       weight  acceleration
mean   25.041637    179.373665   98.715302  2881.505338     15.738790
std     7.912874     95.512897   33.822711   792.548494      2.570191
```

(d) Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.

CODE:

```
sns.pairplot(auto[['mpg','cylinders','displacement','horsepower','weight','acceleration','year']],
diag_kind="kde")
plt.show()
```

OUTPUT:

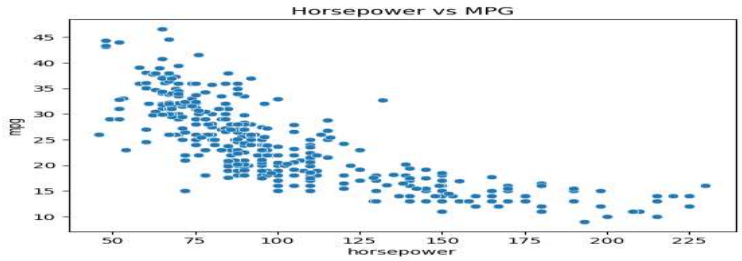


(f) Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg? Justify your answer.

CODE:

```
sns.scatterplot(x='horsepower', y='mpg', data=auto)
plt.title("Horsepower vs MPG")
plt.show()
```

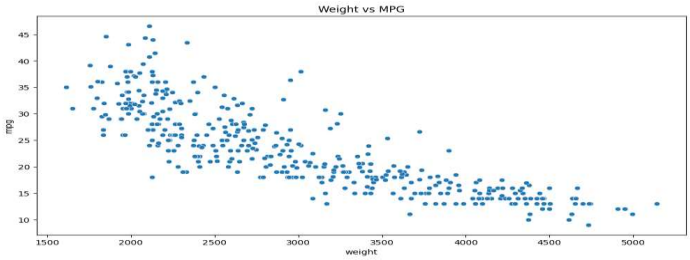
OUTPUT:



CODE:

```
plt.figure(figsize=(12,6))
sns.scatterplot(x='weight', y='mpg', data=auto)
plt.title("Weight vs MPG")
plt.show()
```

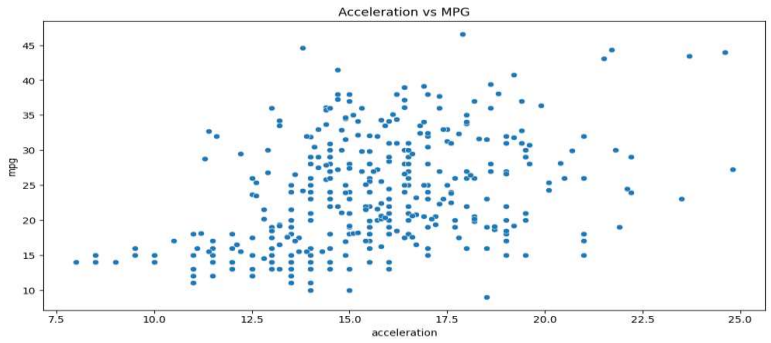
OUTPUT:



CODE:

```
# Acceleration vs MPG scatterplot
plt.figure(figsize=(12,6))
sns.scatterplot(x='acceleration', y='mpg', data=auto)
plt.title("Acceleration vs MPG")
```

OUTPUT:



INTERPRETATION:

- (a) Quantitative predictors: mpg, displacement, horsepower, weight and acceleration. Qualitative predictors: cylinders, year, origin and name (cylinders, year and origin are stored as numbers but behave like categories or levels).
- (b) The predictors sit on very different scales. Ranges (min to max): mpg 9.0 to 46.6, displacement 68 to 455, horsepower 46 to 230, weight 1,613 to 5,140 and acceleration 8.0 to 24.8.
- Means and standard deviations: mpg 23.45 (SD 7.81), displacement 194.41 (104.64), horsepower 104.47 (38.49), weight 2,977.58 (849.40) and acceleration 15.54 (2.76). Weight and displacement have by far the largest spread, so they should be standardised before any scale-sensitive method is used.
- (c) After removing observations 10 to 85, mean mpg rises from 23.45 to 25.04, while mean weight (2,977.6 to 2,881.5), displacement (194.4 to 179.4) and horsepower (104.5 to 98.7) fall; the minimum mpg moves from 9.0 to 11.0. The removed rows held heavier, more powerful, lower-mpg cars, so summary statistics depend on which rows are kept.
- (d) The pairplot shows mpg falling steadily, with some curvature, as weight, displacement and horsepower increase, and these three predictors are strongly positively related to one another. Acceleration has only a weak positive relationship with mpg, and mpg tends to rise with model year.
- (f) Weight, horsepower and displacement are the most useful predictors of mpg because they show the clearest, strongest trends; year is a useful secondary predictor and acceleration a weak one. The strongest predictors overlap heavily (collinearity), which is why a regression on all of them needs to be checked for redundant variables (see Experiment 3).